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BSIT Fall 2021
Software Engineering
Quiz 5

Time Allowed: 15 minutes

Maximum Marks: 20

Explain following relationships with symbols and examples:

1. Containment
2. Dependency
3. Inheritance
4. Association

Answer Space:

1. Containment: It is a kind of relationship in which element encapsulates or includes other element.

It is defined as whole-part relationship. In containment, whole class is somehow associated with parts.

In containment, there can be 2 types of relationships:

- 1) Strong relationship
- 2) Weak relationship

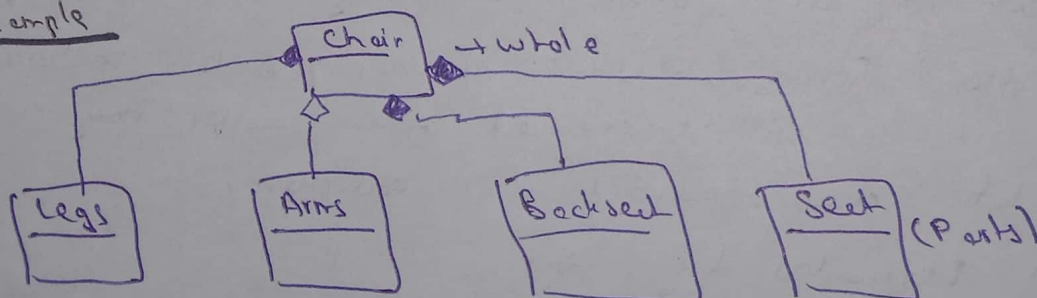
Strong relationship is represented with full diamond

Weak relationship is represented with unfilled diamond

Strong relationship is one which is mandatory without other one part which is of no use... It is also called linked with composition.

Weak relationship is one which is optional. Aggregation is weak relationship.

Example



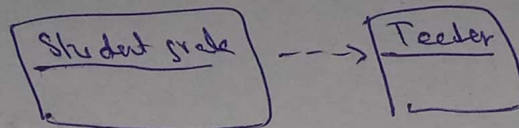
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2. Dependency: Dependency means one class is dependent on the other class. Without one, other can not function properly.. other will only function properly if other class exists.

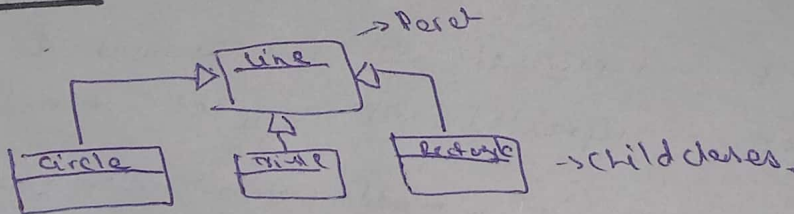
Symbol: Dotted line arrow (--->)

Example



3. Inheritance: Inheritance is kind of relationship between parent and child class. Classes which have inherited properties. Child class inherits properties from parent class.

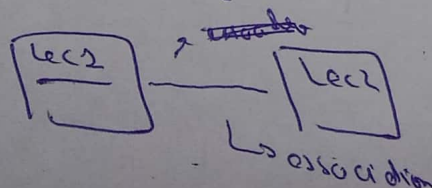
Symbol: Represented by arrow pointed towards parents.



circle, Triangle, rectyls are inherited from line class.

4. Association: Association means that two classes are some how not dependent on each other but they are associated with each other. For eg software engineer and DB lectures are going on. Each lec will be continued after another. They are not dependent but are associated with each other.

Symbol: Represented by (—>)



Lec = lecture. Page 2 of 2